

AIM2 Preparation

Week 2 - Day 6: Seaborn & Advanced Data Visualization

Goal: Use Seaborn with Pandas to create statistical visualizations that help compare categories, inspect distributions, study relationships, and communicate EDA findings.

Day 6 Status: COMPLETE

1. Day Overview

Day 6 builds directly on Day 4 EDA and Day 5 Matplotlib. Instead of only calculating statistics or creating basic plots, Seaborn makes it convenient to create higher-level statistical visualizations directly from Pandas DataFrames.

```
Day 4: EDA / Data Analysis
|
Day 5: Matplotlib Visualization
|
Day 6: Seaborn Statistical Visualization
```

Main tools: Pandas, Matplotlib, Seaborn

2. Seaborn vs Matplotlib

Matplotlib is a fundamental plotting library with detailed control. **Seaborn** is built on top of Matplotlib and provides higher-level functions designed especially for statistical visualization and Pandas DataFrames.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
```

It is normal to use both together. For example, Seaborn can create the plot while Matplotlib controls the title, labels, figure size, rotation, and display.

```
sns.scatterplot(data=df, x="Python", y="AI")
plt.title("Python vs AI")
plt.show()
```

3. Seaborn Bar Plot

A bar plot is useful for comparing a numerical measurement across categories. In the exercise, student names are categories and Average is numerical.

```
sns.barplot(
    data=df,
    x="Name",
    y="Average"
)
plt.title("Student Average Scores")
```

```
plt.xticks(rotation=45)
plt.show()
```

Interpretation: Taller bars represent higher student averages. This makes it easy to visually compare student performance.

4. Histogram and KDE

A histogram shows the distribution of numerical data by grouping observations into intervals called **bins**.

```
sns.histplot(
    data=df,
    x="Math",
    bins=5,
    kde=True
)
plt.title("Math Score Distribution")
plt.show()
```

The key Day 5 idea still applies: fewer bins generally create wider intervals and group more observations together; more bins create narrower intervals and show finer detail.

KDE - Kernel Density Estimate

kde=True adds a smooth curve estimating the shape of the numerical distribution. The KDE is not simply a line connecting the tops of histogram bars; it is estimated from the observations themselves.

```
Histogram bars -> grouped frequency/count
KDE curve      -> smooth estimate of distribution shape
```

5. Box Plot

A box plot summarizes a numerical distribution and is useful for comparing distributions across categories.

```
sns.boxplot(
    data=df,
    x="City",
    y="Math"
)
plt.title("Math Score Distribution by City")
plt.show()
```

Important parts: the middle line is the median (50th percentile); the box represents the middle portion of the data between Q1 and Q3; whiskers extend beyond the box according to the boxplot rule.

```
Q3 / 75%
+-----+
|         |
|-----|  <- Median / 50%
|         |
+-----+
Q1 / 25%
```

In the exercise dataset, the city box plots allow Math score distributions to be compared rather than looking only at a single average.

6. Scatter Plot and hue

A scatter plot is used to inspect the relationship between two numerical variables.

```
sns.scatterplot(
    data=df,
    x="Python",
    y="AI",
    hue="City"
)
plt.title("Python vs AI Scores by City")
plt.show()
```

hue adds another variable as visual grouping. Here, points are grouped by City while Python remains on the x-axis and AI remains on the y-axis.

```
x="Python"    -> numerical x-axis
y="AI"        -> numerical y-axis
hue="City"    -> additional categorical grouping
```

7. Regression Plot

A regression plot combines observed data points with a fitted linear trend line.

```
sns.regplot(
    data=df,
    x="Python",
    y="AI"
)
plt.title("Python vs AI Regression Plot")
plt.show()
```

An upward trend suggests a positive relationship; a downward trend suggests a negative relationship. A regression trend can help visualize association, but it does **not** prove that one variable causes the other.

Core rule: Correlation/association does not imply causation.

8. Correlation Heatmap

A heatmap is a convenient way to visualize a correlation matrix.

```
correlation = df[["Math", "Python", "AI"]].corr()

sns.heatmap(
    correlation,
    annot=True
)
plt.title("Correlation Heatmap")
plt.show()
```

`annot=True` displays the numerical correlation values inside the cells.

The diagonal values are always 1.0 because each variable is perfectly correlated with itself: Math vs Math, Python vs Python, and AI vs AI.

In the Day 6 dataset, the subject correlations are approximately 0.99, indicating very strong positive linear relationships. This still does not establish causation.

9. Pair Plot

A pair plot automatically creates a grid of visualizations for multiple numerical columns, making it useful during EDA.

```
sns.pairplot(
    df,
    vars=["Math", "Python", "AI"],
    hue="City"
)
plt.show()
```

Off-diagonal panels compare pairs of numerical variables. Diagonal panels show each variable's own distribution. Adding `hue="City"` also exposes categorical grouping.

Pairplot Title Improvement

Because `pairplot()` creates an entire `PairGrid`, using a normal `plt.title()` can overlap one subplot. A better whole-grid title is:

```
g = sns.pairplot(
    df,
    vars=["Math", "Python", "AI"],
    hue="City"
)

g.fig.suptitle("Subject Relationships", y=1.02)
plt.show()
```

10. Day 6 Exercises - What You Practiced

Exercise	Visualization	Purpose
1	Bar plot	Compare student Average scores
2	Histogram + KDE	Study Math score distribution
3	Box plot	Compare Math distributions by City
4	Scatter + hue	Study Python vs AI and group by City
5	Regression plot	Visualize Python-AI linear trend
6	Heatmap	Visualize subject correlations
7	Pair plot	Inspect many numerical relationships together

11. Day 6 Assessment - Correct Answers

Q1. Why might Seaborn be easier than Matplotlib with Pandas DataFrames?

Seaborn integrates naturally with Pandas DataFrames and provides higher-level statistical plotting functions, often requiring less code.

Q2. What does hue do?

It maps an additional variable to visual grouping, usually different colors.

Q3. What does the middle line in a box plot represent?

The median, or 50th percentile.

Q4. What does `kde=True` add?

A smooth estimate of the distribution's shape.

Q5. Scatter plot vs regression plot?

A scatter plot shows individual x-y observations; a regression plot also adds a fitted trend line.

Q6. What does `annot=True` do in a heatmap?

It writes numerical values inside the heatmap cells.

Q7. Why are correlation heatmap diagonal values 1.0?

Each variable has perfect correlation with itself.

Q8. Why is pairplot useful for EDA?

It quickly displays multiple pairwise numerical relationships and distributions without manually creating many plots.

Q9. Does a strong upward regression trend prove X causes Y?

No. It indicates association/relationship, not causation.

12. Mini-Project - Student Performance Advanced Visualization

The final Day 6 workflow combined all seven visualization techniques into one student-performance analysis.

Student Performance Advanced Visualization

1. Student Average Scores -> Seaborn bar plot
2. Math Score Distribution -> Histogram + KDE
3. Math Scores by City -> Box plot
4. Python vs AI by City -> Scatter plot + hue
5. Python vs AI Trend -> Regression plot
6. Subject Correlations -> Heatmap
7. Subject Relationships -> Pair plot

This mini-project demonstrates the progression from raw numerical values to statistical visual interpretation.

13. Quick Revision Cheat Sheet

Function	Use
<code>sns.barplot()</code>	Compare numerical values across categories
<code>sns.histplot()</code>	Study numerical distribution/frequency
<code>kde=True</code>	Add smooth distribution estimate
<code>sns.boxplot()</code>	Compare median, quartiles and spread
<code>sns.scatterplot()</code>	Study relationship between two numerical variables
<code>hue=</code>	Add another grouping variable
<code>sns.regplot()</code>	Scatter observations + fitted regression trend
<code>sns.heatmap()</code>	Visualize a matrix such as correlations
<code>annot=True</code>	Display values inside heatmap cells
<code>sns.pairplot()</code>	Automatically compare multiple numerical features

14. Day 6 Completion Summary

You completed Seaborn bar plots, histogram + KDE, box plots, scatter plots with hue, regression plots, correlation heatmaps, and pair plots. You also correctly interpreted positive relationships, median, categorical grouping, correlations, and the difference between association and causation.

Week 2 Day 6 - Seaborn Visualization: COMPLETE

Week 2 Progress

Day 1 - NumPy Fundamentals	COMPLETE
Day 2 - Pandas Fundamentals	COMPLETE
Day 3 - Data Cleaning	COMPLETE
Day 4 - EDA / Data Analysis	COMPLETE
Day 5 - Matplotlib Visualization	COMPLETE
Day 6 - Seaborn Visualization	COMPLETE
Day 7 - Canadian Housing Analysis	NEXT